

Wenjun Cheng

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EDUCATION

University of Michigan (U-M)

B.S.E. in Robotics (GPA: 4.0/4.0)

Ann Arbor, MI

08/2025 – 05/2027 (Expected)

- **Advisors:** Prof. Yulun Tian, Prof. Maani Ghaffari, Dr. Yue Hu

Shanghai Jiao Tong University (SJTU)

B.S.E. in Electrical and Computer Engineering

Shanghai, China

08/2023 – 08/2027 (Expected)

- **Advisor:** Prof. David L.S. Hung

RESEARCH INTERESTS

Spatial Intelligence, Embodied Foundation Models, Long-Horizon Memory, Multi-Agent Systems, Interpretability

RESEARCH EXPERIENCE

Heterogeneous Multi-Robot Collaboration with Vision-Language Models

Ann Arbor, MI

Co-first Author, advisors: Prof. Yulun Tian & Dr. Yue Hu

02/2026 – Present

- Built a system for **heterogeneous multi-robot search** in self-built **Isaac Sim** disaster scenes, in which a **zero-shot VLM coordinator** reasons over each robot's camera height, speed, and terrain-dependent motion cost to assign tasks from a **shared spatial map** fused across the team. Accepted to two RSS 2026 workshops.
- Extending the project into a **benchmark of interdependent long-horizon missions**, such as a humanoid opening a door to retrieve a medical kit and loading it onto a quadruped that delivers it through rough terrain. Built its embodiment layer with **GEAR-SONIC** on the Unitree G1 and an **Extreme Parkour** policy adapted to the Go2 with a redesigned terrain curriculum.
- Building an **agentic baseline** in which each robot's **VLM agent** acts through **tool calls**, maintains a **spatio-temporal memory**, and coordinates via a **captain-led planning protocol**.

Probing and Training Ego-Motion Understanding in Navigation-Trained VLMs

Ann Arbor, MI

Project Lead, advisors: Prof. Yulun Tian & Dr. Yue Hu

01/2026 – Present

- Asked whether navigation training teaches VLMs to **estimate their own motion**, comparing Qwen3-VL before and after navigation fine-tuning (LongNav-R1) with **SE(3) motion probes** on a **controlled dataset** of single-motion image pairs.
- Found that training **sharpens the internal ego-motion estimate**, even for motions the policy never takes, **yet the model's answer never improves**, and traced the estimate to **cross-view attention**. Now testing the reverse direction, whether ego-motion supervision improves navigation.

Task-Conditioned Active Semantic Information Acquisition

Ann Arbor, MI

Research Assistant, advisors: Prof. Yulun Tian & Prof. Vasileios Tzoumas

06/2026 – Present

- Leading the **Unreal Engine** simulation for a **task-conditioned active perception** framework, in which robots gather observations to resolve **semantic hypotheses** by propagating uncertainty from an **object-level map belief**, starting with a single robot and extending to multi-robot teams.

Structured Spatial Memory for Vision-and-Language Navigation

Ann Arbor, MI

Research Assistant, advisors: Prof. Maani Ghaffari & Dr. Yue Hu

10/2025 – 02/2026

- Evaluated **scene graphs and semantic memory** for VLM agents in long-horizon VLN, finding that memory alone cannot resolve multi-view ambiguity and tracing the failure to weak **VLM motion grounding**, which motivated the ego-motion probing project.

- Designed a **liquid-impact platform** with vibration isolation and tunable impact speed, angle, droplet size, and liquid type.
- Built a synchronized **multimodal acquisition pipeline** coupling piezoelectric force sensing, high-speed imaging, and an infrared trigger circuit, with **FFT-based** analysis that separates impact signals from structural noise.

PUBLICATION

[1] **HeRo-Nav: Heterogeneous Multi-Robot Collaboration for Semantic Navigation with Vision Language Models**
Siddhant Tandon*, **Wenjun Cheng***, Junzhe Wu, Yue Hu†, Yulun Tian†
Open World Navigation in the Foundation Model Era: Robustness and Failure Recovery & Robot Planning in the Era of Foundation Models, RSS 2026 Workshop.

PROFESSIONAL EXPERIENCE

Edan Instruments, Inc.

Shenzhen, China

Algorithm Intern, R&D System Department

01/2025

- Prototyped the company’s first **interactive 3D heart visualization** for ECG diagnoses, mapping diagnostic labels to cardiac regions on an adapted **Blender** model. Approved for production integration.

TEACHING EXPERIENCE

SJTU Global College

Shanghai, China

Tech Teaching Assistant (TA) for ENGR1000J Introduction to Engineering

09/2024 – 12/2024

- Co-designed a track race challenge for ultrasonic-guided robotic cars and mentored about 40 freshmen through it and a follow-up open-ended build, teaching C++, SolidWorks, and Arduino. Nominated for the **Annual Outstanding TA Award**.

LEADERSHIP

SJTU Global College Tennis Association

Shanghai, China

Founding President

08/2024 – 08/2025

- Founded the institute’s tennis club, growing it to 100+ members and organizing 10+ intra- and inter-institute competitions.

TECHNICAL SKILLS

Robotics	ROS 2, Isaac Sim, Unreal Engine, Habitat, MuJoCo, SLAM
Machine Learning	PyTorch, Vision-Language Models
Programming	Python, C++, C, MATLAB, Verilog
Tools	Git, Linux, Docker, LaTeX, SolidWorks, Blender

COURSEWORK & HONORS

Graduate Coursework	Foundation of Language Models (A), Mobile Robotics (A+), Multi-Robot Systems (A), Self-Driving Cars (in progress)
Honors (U-M)	University Honors; Dean’s List; Michigan Engineering Continuing Student Scholarship
Honors (SJTU)	Huatai Securities Technology Scholarship; Yu Liming Scholarship; Outstanding Undergraduate Scholarship